



Contents lists available at ScienceDirect

Advanced Exercise and Health Science

journal homepage: www.keaipublishing.com/en/journals/aehs

Decoding exercise-induced delayed aging from a single-cell perspective

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ARTICLE INFO

Keywords:

Aging
Exercise
Skeletal muscle
Nervous system
Single-cell omics

ABSTRACT

Aging is a systematic process characterized by the functional decline of multiple organs such as skeletal muscle, the nervous system, and the cardiovascular system. Age-related conditions such as sarcopenia, neurodegenerative diseases, and vascular dysfunction have become major health challenges in aging societies. Regular physical activity, a safe and effective non-pharmacological intervention, has been extensively demonstrated to ameliorate aging-related phenotypes across these systems. Recent advances in single-cell omics technologies provides an unprecedented tool to dissect the precise cellular regulatory networks underlying exercise interventions. This review systematically summarizes the recent progress in applying single-cell omics to decode how exercise delays multi-organ and systemic aging. Findings reveal that the aging process exhibits significant cellular heterogeneity, which is manifested by abnormal function of muscle stem cells, neural precursor cells, and specific endothelial cell subsets, accompanied by the formation of a chronic inflammatory microenvironment and dysregulation of intercellular communication networks. Exercise exerts its anti-aging effects by remodeling the immune landscape, repairing stem cell niches, and regulating key signaling pathways including notch, transforming growth factor-beta, and myokines networks. In summary, single-cell omics has enabled preliminary mapping of the spatiotemporal molecular networks and cellular heterogeneity through which exercise delays multisystem aging, offering a new paradigm for interventions against age-related diseases and revolutionizing our understanding of exercise-mediated aging regulation. These advances lay a solid foundation for the rational design of precision exercise interventions and novel therapeutic strategies to mitigate age-related functional decline and foster healthy aging.

1. Introduction

As the global population ages at an accelerating pace, achieving "healthy aging", delaying functional decline, and preventing age-related diseases have become core challenges in the fields of public health and life sciences in the 21st century. Aging is a complex biological process. At the molecular and cellular levels, it is characterized by a series of hallmarks, including genome instability, telomere attrition, epigenetic alterations, loss of proteostasis, impaired autophagy, dysregulated nutrient sensing, mitochondrial dysfunction, cellular senescence, stem cell exhaustion, altered intercellular communication, chronic inflammation, dysbiosis, extracellular matrix

(ECM) remodeling, and psychosocial isolation.¹ Collectively, these fundamental alterations drive imbalances in tissue cellular composition and functional dysregulation.² Beyond the aforementioned cellular and molecular features, the macroscopic manifestations of aging include the progressive decline of multiple organ functions, impaired homeostatic maintenance, and a marked increase in disease susceptibility.¹ These changes are associated with reduced physiological function and underlie a spectrum of disorders, including cardiovascular, cardiopulmonary, musculoskeletal, metabolic, and cognitive diseases.³

Among numerous intervention strategies, regular physical activity is regarded as one of the most effective and safest non-pharmacological

Peer review under responsibility of Wuhan Sports University.

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Received 28 January 2026; Received in revised form 30 May 2026; Accepted 31 May 2026

Available online 8 June 2026

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anti-aging interventions, and is often hailed as a "pleiotropic drug".^{4,5} A wealth of clinical and basic studies has confirmed that exercise systematically improves metabolic health, enhances cardiovascular function, boosts cognitive performance, preserves muscle mass and strength, and modulates immune function, thereby reducing the incidence of multiple chronic diseases and comprehensively enhancing the quality of life in older adults.^{6–8} However, a core, yet incompletely elucidated, scientific question remains: What are the fundamental cellular and molecular origins underlying these global benefits of exercise? Traditional bulk RNA-seq, a bulk tissue profiling technique, averages the signals of thousands of heterogeneous cells, masking the unique or even opposing changes in specific cell subtypes and hindering our ability to identify the precise targets of exercise intervention within complex tissue microenvironments.

In recent years, revolutionary advances in single-cell omics have provided unprecedented resolution for dissecting the complexity of biological processes. Techniques such as single-cell RNA sequencing (scRNA-seq) and single-cell assay for transposase-accessible chromatin sequencing (scATAC-seq) enable the construction of high-resolution "cellular atlases" for tissues or organs, allowing the identification of unknown or rare cell types, the characterization of continuous dynamic transitions in cell states, and the delineation of intercellular communication networks.⁹ As single-cell omics technologies continue to evolve, we can assess the state and function of individual cells within tissues across multiple layers, opening a new avenue for future experimental investigations. The fields of single-cell omics and aging are now increasingly intertwined. As will be discussed, aging exerts distinct effects on different cell types, providing a direct rationale for employing single-cell approaches to track aging-related changes.

Therefore, this review aims to systematically review and integrate how to use the powerful tool of single-cell omics to analyze the deep mechanisms of exercise against aging. We will first focus on how single-cell technology research can reveal aging-related degenerative changes in the cellular landscape in key organ systems such as skeletal muscle, nervous system, and cardiovascular systems, and how exercise can restore tissue homeostasis by reshaping these cellular landscapes. Secondly, from a single-cell perspective, we will briefly discuss the systemic effects of exercise on multiple tissues and how exercise remodels intercellular and inter-organ communication networks. Finally, we will discuss the challenges facing this field and look forward to the grand prospect of integrating analysis and causality research across organs in the future, ultimately providing a theoretical basis for achieving personalized, precise exercise programs based on cellular and molecular information.

2. Aging and cellular senescence

Aging refers to the process by which an organism's physiological functions gradually decline with advancing age.⁵ With increasing age, diverse types of stress induce cellular senescence and contribute to the development of age-related diseases, including cardiovascular diseases, diabetes, Alzheimer's disease, osteoarthritis, and cancer.^{10–13} Senescent cells are critical drivers of organismal aging, and the accumulation of senescent cells is considered one of the fundamental underlying mechanisms of aging.² Cell cycle arrest is a hallmark prerequisite for cellular senescence.¹⁴ Based on distinct inducing factors, cellular senescence is mainly classified into four subtypes: replicative senescence (RS)¹⁵, oncogene-induced senescence (OIS), stress-induced premature senescence (SIPS), and treatment-induced senescence (TIS).^{16–18} Different senescence subtypes have distinct pathophysiological and biological significance, including their roles in chronic diseases and aging-related disorders. For example, OIS acts as a safeguard against cancer by preventing the proliferation of potentially malignant cells.^{19,20} TIS contributes to tissue repair after injury, but if unresolved, it may lead to fibrosis.²¹ SIPS, exemplified by reactive oxygen species (ROS)-induced senescence, protects against damage but can result in chronic

inflammation and tissue dysfunction if senescent cells accumulate.²² RS represents the natural aging process of cells, leading to a decline in tissue function and regenerative capacity.²³

Among these, RS and SIPS are the two most common and biologically significant subtypes in the context of natural aging, as they are directly associated with age-related physiological decline.¹⁵ Therefore, these two subtypes are elaborated in detail below. RS is associated with the progressive loss of telomeric DNA during each round of cell division.²⁴ Owing to the 5' to 3' directionality of DNA polymerases, the DNA replication machinery cannot copy the ends of linear chromosomes.⁵ Over time, this progressive telomeric DNA loss ultimately triggers cell cycle checkpoints that halt further cell division.⁵ This form of RS, also known as the Hayflick limit, occurs in numerous species and is linked to species-specific maximum lifespan.¹⁴ Accordingly, this gave rise to the telomere theory of aging.² When cellular damage is irreparable, cells undergo senescence rather than programmed cell death.²⁵ SIPS, in contrast to RS, is triggered by sublethal exposure to various stressors without requiring critical telomere shortening.²⁶ SIPS was defined using various cellular stresses, such as ultraviolet radiation (UV), hydrogen peroxide (H₂O₂), chemotherapeutic agents, and ionizing radiation (IR), that caused premature senescence phenotypically similar to replicative senescence.²⁷ Sub-cytotoxic doses of these stresses are required to efficiently induce SIPS.²⁸ Many cell types—including lung fibroblasts, skin fibroblasts, embryonic fibroblasts, melanocytes, endothelial cells, and retinal pigment epithelial cells—undergo SIPS upon these treatments.²⁸ Interestingly, SIPS, rather than apoptosis, is the major cellular response in these cell types both *in vitro* and *in vivo*.²⁴ Cells undergoing SIPS share many cellular and molecular features with cells undergoing replicative senescence, including senescence-associated β -galactosidase (SA- β -gal) activity, senescence-associated secretory phenotypes (SASP), and upregulation of p16/p21.²⁹ The primary difference between RS and SIPS lies in the timing and programming of these features: RS is programmed to occur at a specific time when telomeric DNA ends become exposed, whereas SIPS is not programmed but rather a responsive reaction to a given stress.²⁹ SIPS is not triggered by dysfunctional telomeres.³⁰ A common feature shared by both types of senescence is the permanent activation of DNA-damage checkpoint machineries.^{31,32} The distinction is that irreparable damage at telomere-nonspecific sites serves as the activating signal for SIPS, while specific telomeric ends serve as the activating signals for RS.³² Importantly, the aging microenvironment *in vivo*—characterized by chronic low-grade inflammation, elevated ROS, and accumulated DNA damage—closely resembles the conditions that induce SIPS.³³ Consequently, SIPS is increasingly recognized as a more representative model of *in vivo* cellular senescence during natural aging than RS.²⁴

Senescent cells are non-dividing cells characterized by aberrant secretory phenotypes, cellular damage, and metabolic reprogramming, and they frequently evade apoptosis.³⁴ Molecular identification of senescent cells remains challenging, as the senescent state induced by distinct triggers in different tissues exhibits substantial heterogeneity.³⁵ Nevertheless, key markers for cellular senescence identification include SA- β -gal activation, SASP, and elevated expression of two cyclin-dependent kinase inhibitors (CDKN2A encoding p16, CDKN1A encoding p21).³⁶ Organ aging is associated with the progressive deterioration of tissue structure and organ function, leading to organ dysfunction and elevated risk of chronic diseases such as neurodegenerative disorders (NDDs), cardiovascular diseases (CVDs), and sarcopenia.³⁷ Experimental evidence suggests that the decline in organismal physiological functions is primarily driven by cellular and organ aging.³⁸ Therefore, exploring cellular senescence and studying the dynamics of different cell types during aging have become a top priority for assessing age-related changes, tracking physiological aging processes, and predicting pathological conditions. To date, with the advancement of single-cell omics technologies, researchers have gained unprecedented insights into alterations in the cellular composition of various organs across the lifespan. These studies offer promising strategies for attenuating the aging process and reducing the risk of age-related diseases.

3. Single-cell decoding of exercise remodeling in aged skeletal muscle

3.1. Single-cell atlas of aging skeletal muscle

Skeletal muscle accounts for 40% of human body mass and plays a pivotal role in metabolism and immune regulation.^{39–41} It is mainly composed of multinucleated muscle fibers (MFs), which can be classified into slow-twitch muscle fibers (Type I) and fast-twitch muscle fibers (Type IIA, Type IIX, and intermediate hybrid fibers) based on contraction speed, structural protein composition, and metabolic activity.⁴² The contraction of MFs is controlled by the activity of motor neurons that innervate muscle fibers at the neuromuscular junction.^{43,44} MFs are surrounded by mononuclear muscle stem cells (MuSCs), which can generate new MFs following injury.⁴⁴ In addition, the muscle microenvironment comprises supportive fibroblasts, blood vessels, immune cells, Schwann cells, and nerve axons.⁴⁵

Given its close interconnection with other body tissues, skeletal muscle is a critical determinant of overall health.^{46,47} Loss of muscle mass and strength is a hallmark of skeletal muscle aging, often leading to sarcopenia.⁴⁸ Sarcopenia is the leading cause of falls and fractures in older adults and the second leading cause of injury and mortality in this population.⁴⁹ Therefore, it is of great importance to explore the effects of aging on the alterations of key cell types in skeletal muscle and the modulatory effects of exercise on aged skeletal muscle using single-cell omics technologies.

In mammals, the stem cell niche is a key factor in maintaining quiescence—a reversible state of growth arrest that is essential for preserving the number and function of adult stem cells.^{50,51} Skeletal muscle homeostasis and regeneration rely on a population of muscle-specific, *Pax7*-expressing adult stem cells termed satellite cells, which are typically quiescent.⁵² Upon stimulation by injury or stress, quiescent satellite cells in the G0 phase become activated, re-enter the cell cycle to proliferate, and differentiate into new muscle fibers, or replenish the satellite cell pool via self-renewal.^{53,54} In previous heterochronic parabiosis and xenograft experiments, researchers proposed that satellite cell-dependent muscle regeneration declines with aging and can be reversed by exposure to a youthful systemic environment.^{55–59} In mouse models, Pedro et al.⁶⁰ demonstrated that the “reversible quiescence” maintained by mammalian MuSCs depends on active epigenetic silencing of aging-related genes. With advancing age, the level of histone 2A ubiquitination mediated by the polycomb repressive complex 1 decreases, leading to de-repression of *CDKN2A* (encoding *p16INK4a*), which in turn drives the transition of stem cells from a quiescent state to a pre-senescent state.⁶⁰ This transition is irreversible and ultimately results in stem cell pool exhaustion and loss of muscle regenerative capacity.⁶⁰ However, whether this irreversible transition occurs in human MuSCs—and whether *p16INK4a* upregulation similarly drives irreversible senescence in human cells—remains to be determined.⁶⁰ Furthermore, multiple scATAC-seq studies in mice have confirmed that skeletal muscle aging is accompanied by profound chromatin accessibility changes linked to alterations in muscle fiber composition and myonuclear characteristics.^{61–64} Specifically, the number of quiescent myogenic stem cells in aged muscle decreases, while the number of non-myogenic tissue-resident cells increases correspondingly.^{62,65} Surviving myogenic stem cells persistently exhibit aberrant activation of inflammation- and stress-related signaling pathways, leading to impaired proliferation and differentiation.^{61,62} This chronic activation is associated with accelerated stem cell depletion in mouse models, raising the possibility that it partially explains the decline in skeletal muscle regenerative capacity in older adults.⁶⁰ However, direct causal evidence in humans is lacking, as most studies are correlative and cannot distinguish cause from effect.⁶²

A single-cell and single-nucleus transcriptomic atlas of human skeletal muscle was generated from 31 donors aged 15–99 years (adults: 15–46 years, *n* = 12; elderly: 74–99 years, *n* = 19; 17 males, 14

females).⁶² Profiling over 387,000 cells and nuclei, this atlas revealed that the heterogeneity of MuSCs increases with age.⁵⁷

In addition, the decline in skeletal muscle mass and strength with age is mainly attributed to the selective reduction in the number and size of fast-twitch multinucleated MFs. Several single-cell studies have shown that fast-twitch MFs are more susceptible to age-related atrophy than slow-twitch MFs.^{44,62} During aging, Type I myonuclei shift toward a more glycolytic phenotype via metabolic reprogramming.⁶⁶ In contrast, Type II myonuclei display increased glycogen consumption and protein catabolic processes, explaining their higher susceptibility to atrophy.⁴⁴ Fast-twitch Type IIX fibers in the intercostal muscle are almost completely lost with age, yet this loss is not observed at the nuclear level.⁴⁴ This apparent paradox arises from the emergence of fast-Type IIX myonuclei in both slow-twitch and fast-twitch Type IIA fibers with advancing age.⁶²

Regarding the muscle microenvironment, single-cell studies in mice have shown upregulated expression of pro-inflammatory chemokine genes in stromal cells such as muscle fibroblasts, which may mediate aberrant recruitment of lymphocytes and collectively contribute to the establishment and maintenance of a chronic inflammatory microenvironment in aged muscle.⁵² Specifically, a single-cell study analyzing hindlimb muscles from five young (3 months) and three aged (19 months) mice sequenced 68,956 cells and 27,573 nuclei, revealing these age-associated inflammatory changes.⁶¹ Concurrently, the cell state of adipogenic progenitor cells undergoes a dramatic shift, from a regenerative-supportive phenotype to a pro-fibrotic one.⁶⁷ This transition is often accompanied by the accumulation of mature adipocytes, which may further increase the risk of age-related phenotypic alterations in muscle tissue.⁶⁷ In humans, droplet-based 3' sequencing was performed on intercostal muscle biopsies from eight young (approximately 20–40 years) and nine aged (approximately 60–75 years) donors, generating transcriptomic profiles of 90,902 cells and 92,259 nuclei.⁶⁷ Single-cell studies in humans have also revealed upregulation of the pro-inflammatory chemokine gene *C-C motif chemokine ligand 2* (*CCL2*) (encoding *CCL2*) in fibroblasts, MuSCs, arterial endothelial cells, smooth muscle cells (SMCs), and perivascular cells during aging.⁵⁵ In addition to this pan-microenvironmental upregulation of *CCL2* (encoding *CCL2*), increased expression of *CCL3* (encoding *CCL3*), *CCL4* (encoding *CCL4*), and *C-X-C motif chemokine ligand 8* (*CXCL8*) (encoding *CXCL8*) has also been observed during aging.⁵¹ Collectively, in both mice and humans, skeletal muscle aging is characterized by global downregulation of gene expression, increased inflammation, and reduced activity of pathways involved in growth, repair, and innervation.⁶¹ Notably, the pan-microenvironmental upregulation of *CCL2* with age observed in mice is not recapitulated in humans, indicating species-specific differences in inflammatory coordination.⁶²

Furthermore, a study using combined single-cell and single-nucleus transcriptomics found that macrophages are the most abundant cell type in the skeletal muscle immune microenvironment, which is composed of diverse innate and adaptive immune cells.⁶¹ As the primary immune cell type in injured muscle, macrophages play a critical role in the initiation and resolution of inflammation and significantly promote muscle repair.^{44,68,69} In response to anti-inflammatory cytokines, M1 macrophages transition to M2 macrophages during the later stages of skeletal muscle injury repair.⁷⁰ Subsequently, these M2 macrophages, together with resident M2 macrophages, promote the repair and regenerative processes by enhancing myoblast differentiation and angiogenesis, and stimulating fibro-adipogenic progenitors (FAPs) to produce ECM.⁷¹ However, macrophages in aged muscle tend to polarize toward a pro-inflammatory phenotype, thereby exerting a negative impact on the repair and regenerative capacity of injured muscle.⁷² Currently, targeting immune cell-related pathways in the microenvironment has emerged as an important therapeutic strategy for sarcopenia. Further understanding of the unique functions and origins of macrophage subsets will provide valuable insights for the development of therapies targeting specific pro-inflammatory or fibrotic phenotypes.

In addition, the reduction in acetylcholine receptor-associated cell types and terminal Schwann cells at the neuromuscular junction collectively indicates functional and morphological degenerative changes of this structure with age.⁶¹ The single-cell studies described above, most of which were conducted in mice, collectively reveal a complex network of disrupted intercellular communication in aged skeletal muscle. MuSCs receive aberrant inflammatory signals from pro-inflammatory macrophages and senescent fibroblasts, which suppress their self-renewal capacity.⁷¹ Macrophages themselves exhibit an age-related shift toward a pro-inflammatory phenotype, further amplifying local inflammation.⁷³ FAPs, under the influence of senescent cell-derived SASP factors, transition from a regenerative-supportive to a pro-fibrotic state.⁷⁴ Muscle fibers undergo metabolic reprogramming and type switching.⁷³ The senescent phenotypes of these multiple cell types are interrelated and act synergistically: the sustained release of SASP factors in the local skeletal muscle creates a chronic inflammatory microenvironment, which induces the decline of MuSCs self-renewal capacity, muscle fiber atrophy, and aberrant activation of FAPs leading to fibrosis.⁷⁵ Disrupted intercellular communication further exacerbates the blockade of regenerative pathways and metabolic homeostasis imbalance, ultimately driving the progression of sarcopenia and related muscle dysfunction (Fig. 1). However, most current single-cell and single-nucleus studies only describe these communication disruptions at a correlative level, and functional validation through cell-type-specific pathway perturbations or intercellular signaling blockade is largely lacking. Future studies integrating single-cell transcriptomics with spatial transcriptomics or ligand-receptor interaction modeling will be essential to map the directionality and causality of these communication networks. Furthermore, it remains unclear whether these age-

related changes are primarily attributable to intrinsic alterations in gene expression within MFs or to changes in cell populations within the microenvironment. Notably, through single-cell transcriptomic studies and explorations in skeletal muscle, researchers have demonstrated that targeted inhibition of *p16INK4a* can effectively reverse the senescent phenotype of stem cells and restore their regenerative function in aging and progeria mouse models.⁷⁶ In muscle tissue from elderly sarcopenia patients, *p16INK4a* expression levels are also elevated, suggesting that this mechanism may be relevant in humans.⁷⁷ However, whether *p16INK4a* inhibition would produce similar functional benefits in humans—and whether such an approach would be safe given *p16INK4a*'s tumor suppressor role—requires further investigation in human clinical trials. Nevertheless, these findings provide a potential therapeutic direction for clinical intervention in age-related muscle decline.

When interpreting these findings using single-cell and single-nucleus technologies, several caveats should be noted. First, the sample sizes of the studies discussed above are relatively small. For example, the Lai et al.⁶² Study included only 31 human samples (12 adults, 19 elderly), and the mouse studies included only 8 animals (5 young, 3 middle-aged). This limits statistical power and precludes detailed investigation of confounding biological covariates such as sex, body mass index (BMI), or physical activity levels. Second, current single-cell and single-nucleus studies in skeletal muscle have not separately analyzed sex-specific differences, despite including both male and female participants. Third, the aged mice used in most studies are approximately 19 months old, which is generally considered middle-aged rather than aged, although they already exhibit some senescence-associated features. Moreover, human studies have included only a small number of middle-aged donors (approximately 50–60 years). These experimental

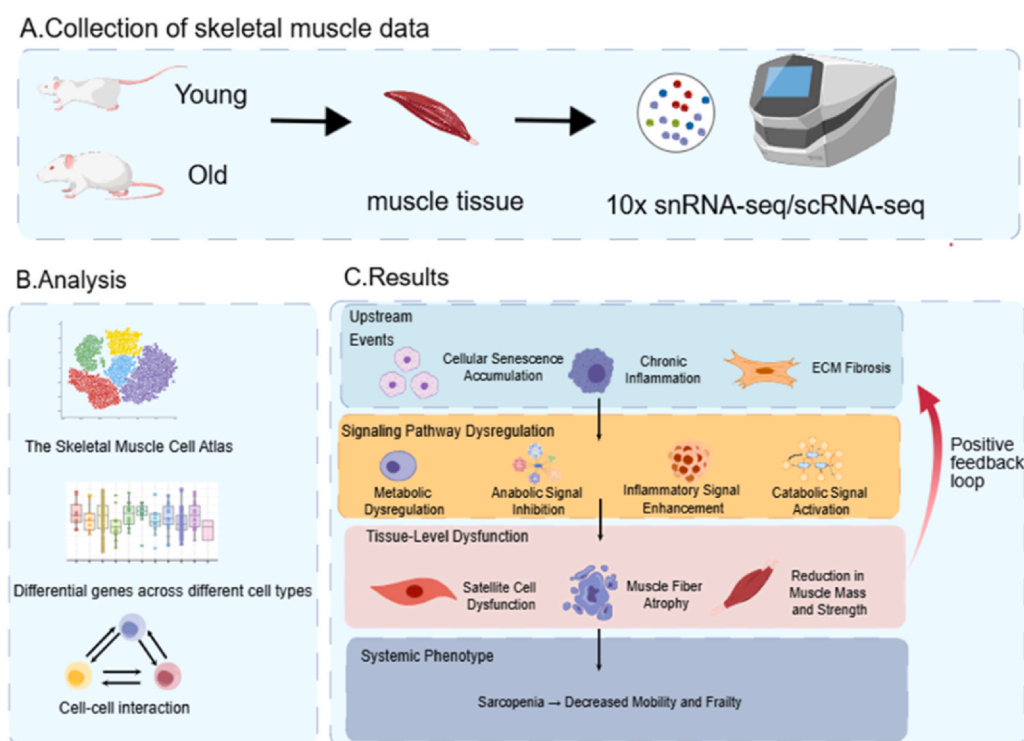


Fig. 1. Schematic overview of the multi-step pathological cascade underlying aging-related skeletal muscle degeneration and sarcopenia progression. (A) Study workflow: Skeletal muscle tissues are harvested from young and aged mice, followed by single-nucleus/single-cell RNA sequencing (10 × snRNA-seq/scRNA-seq). (B) Bioinformatic analysis pipeline: Construction of the skeletal muscle cell atlas, identification of cell-type-specific differentially expressed genes, and evaluation of intercellular communication networks. (C) Mechanistic cascade of aging-driven sarcopenia. Upstream aging-related events, including cellular senescence accumulation, chronic inflammation, and ECM fibrosis, construct a deleterious pro-senescent tissue microenvironment. This microenvironment further induces widespread signaling pathway dysregulation, featured by metabolic dysfunction, suppression of anabolic signaling, amplified pro-inflammatory signaling, and hyperactivation of catabolic signaling. These molecular aberrations subsequently drive tissue-level functional defects: satellite cell dysfunction (blunted muscle regenerative capacity), skeletal muscle fiber atrophy, and progressive loss of muscle mass and contractile strength. Ultimately, these pathological alterations culminate in the systemic clinical phenotype of sarcopenia, accompanied by reduced physical mobility and increased frailty. Notably, these pathological changes also form a self-reinforcing positive feedback loop to exacerbate disease progression. (created with biogdp.com).

material limitations restrict the ability to map continuous aging trajectories in skeletal muscle using single-cell technologies. Fourth, the cross-sectional design of all these studies precludes causal inference; the observed associations between chemokine upregulation and aging do not establish causality. Fifth, species-specific differences in inflammatory coordination - particularly the discordant *CCL2/Ccl2* expression patterns between humans and mice noted above - highlight the need for caution when extrapolating mouse-derived findings to human aging. Currently, no studies have directly compared mouse and human skeletal muscle at the single-cell or single-nucleus level. Future efforts incorporating multi-cohort and multi-species data with larger sample sizes, broader age ranges (including continuous sampling from young to old age), and better-controlled covariates will be essential to clarify the extreme biological differences between young and aged states, as well as the translational relevance of mouse models to human skeletal muscle aging.

3.2. Single-cell omics reveals mechanisms of exercise intervention in aging skeletal muscle

Numerous traditional studies have confirmed that regular physical exercise can effectively reduce the burden of cellular senescence, decrease the age-related abnormal accumulation of senescent cells in tissues, and significantly improve the metabolic function and regenerative potential of skeletal muscle.⁷⁵ However, these studies were largely conducted at the level of whole muscle or mixed MFs, providing limited fine-scale information on how exercise affects distinct cell types within skeletal muscle. Existing animal and human studies have demonstrated that many cell populations in skeletal muscle - including immune cells, mesenchymal cells, MuSCs, and endothelial cells - are essential for a favorable adaptive response to exercise.^{78,79}

In recent years, the application of scRNA-seq has provided new perspectives for dissecting this complexity. An exploratory human study employed scRNA-seq to analyze the effects of acute high-intensity interval exercise on skeletal muscle cells.⁷⁵ This study recruited three healthy, moderately active individuals, including two 26-year-old young participants (one female, one male) and one 50-year-old middle-aged/older participant (female).⁸⁰ The subjects completed an acute exercise protocol consisting of three 30-second all-out sprints (i.e., Wingate tests) separated by 2 min of rest, and skeletal muscle biopsies were collected after exercise for sequencing analysis (the specific post-exercise time point for biopsy collection was not explicitly reported in the study).⁸⁰ The results showed that although a single bout of acute exercise did not significantly alter the relative abundance of major cell populations (except for an increase in lymphocyte numbers, primarily attributed to post-exercise skeletal muscle hyperemia), it induced robust, cell type-specific transcriptional reprogramming.⁸⁰ The transcriptional responses varied dramatically across cell types, with mesenchymal cells exhibiting the most pronounced changes.⁸⁰ Their differentially expressed genes were enriched in ECM remodeling pathways; endothelial cells and monocytes also showed strong responses, while three subpopulations of myogenic cells exhibited a trend toward differentiation into slow-twitch or fast-twitch muscle fibers following exercise stimulation.^{80,81} This suggests that dynamic ECM remodeling is a core component of the muscle adaptive response and that multiple cell types participate in this process in a highly heterogeneous and coordinated manner.

The following subsections will elaborate on the cellular and molecular mechanisms of exercise intervention in aging skeletal muscle from four perspectives: remodeling of the immune microenvironment, restoration of intercellular communication, regulation of MFs and myokines, and functional modulation of MuSCs subpopulations.

3.2.1. Remodeling of the immune microenvironment

The primary regulatory effect of exercise on aged skeletal muscle is the remodeling of immune microenvironment homeostasis, which is

highly consistent with the core conclusion that physical exercise reduces the burden of senescent cells and decreases CS accumulation.⁶¹ In animal models, Liu et al.⁶⁷ demonstrated via scRNA-seq analysis that long-term voluntary aerobic exercise significantly reverses age-associated inflammatory imbalance in skeletal muscle. In this study, aged (22-month-old) male mice were used as the aging model, with young (4-month-old) male mice serving as controls.⁶⁷ Exercise group mice were housed in cages equipped with freely rotating running wheels (connected to digital recorders to monitor running distance).⁶⁷ After a one-week acclimation period, mice performed voluntary wheel running during the dark phase of the sleep-wake cycle for 5 weeks.⁶⁷ Control mice were housed in identical cages with the running wheels removed.⁶⁷ Skeletal muscle samples were collected after 5 weeks of exercise for scRNA-seq analysis.⁶⁷ The results revealed that long-term voluntary exercise significantly downregulated the infiltration of pro-inflammatory macrophages (*CD11b*⁺, *Ly6C*⁺) and neutrophils in aged mouse skeletal muscle, increased the proportion of *CD206*⁺ anti-inflammatory macrophages (M2 phenotype), and restored the physiological number of muscle-resident *CD206*⁺ monocytes, thereby correcting aging-induced immune infiltration abnormalities at the cellular compositional level.⁶⁷ At the molecular level, exercise was associated with suppressed gene transcription and protein secretion of pro-inflammatory pathways such as tumor necrosis factor- α (TNF- α) and interferon- γ (IFN- γ), as well as upregulated expression of anti-inflammatory molecules such as *ANXA1*, effectively reducing the global inflammatory score of skeletal muscle.⁸² It is worth noting that this study employed a long-term chronic exercise paradigm (5 weeks of voluntary wheel running),⁶⁷ which differs fundamentally from the single acute exercise protocol (3 $\times$ 30-second sprints) used in the aforementioned human study in terms of exercise type, intensity, duration, and biological effects.⁶⁷ Furthermore, this study included only male mice and did not explore sex differences in exercise responses.⁶¹ Additionally, while the comparison between aged (22 months) and young (4 months) animals revealed age-related effects, it did not include transitional age groups such as middle-aged animals.

3.2.2. Repair of intercellular communication

Exercise remodels the cellular communication network within the MuSCs niche, repairs aging-induced signaling impairments, and creates a favorable microenvironment for the proliferation and differentiation of MuSCs, thereby synergizing with the ECM remodeling regulated by mesenchymal cells.⁶⁷ Meanwhile, single-cell studies of aerobic exercise have revealed that after five weeks of wheel running, mice restore the signaling capacity of fibro-adipogenic progenitors, Schwann cells, and endothelial cells, enhance the efficiency of ligand-receptor interactions between macrophages and MuSCs, and restore the paracrine signaling activity of key pathways such as notch and Transforming Growth Factor- β (TGF- β).⁶⁷

As a physical vehicle for intercellular communication, the ECM undergoes remodeling mediated by exercise-induced transcriptional reprogramming of mesenchymal cells.⁸³ This reprogramming drives the degradation of damaged ECM components by upregulating the expression of matrix metalloproteinases (MMPs) and promotes the synthesis of new ECM proteins, thereby providing structural support for intercellular signal transduction.⁸⁴

This dual regulation of signaling pathway repair and ECM remodeling effectively optimizes the microenvironment for MuSCs proliferation and differentiation, and mitigates aging-induced defects in MuSCs activation.

3.2.3. Regulation of muscle fibers and muscle factors

As the functional core of skeletal muscle, exercise-induced fiber-specific regulation is essential for muscle regeneration and metabolic improvement.⁸⁵ Exercise significantly upregulates the expression of secreted phosphoprotein 1 (*SPP1*) in muscle fibers, and ELISA validation has confirmed that *SPP1* levels are significantly elevated in both

skeletal muscle tissue and peripheral plasma following exercise.^{67,86} By binding to integrin receptors on the surface of various muscle cell types, *SPP1* reverses the aberrant expression of aging-related target genes.⁸⁷ On one hand, it inhibits the release of SASP factors such as interleukin-1 beta (*IL1B*) and *CCL3*, thereby alleviating local inflammation; on the other hand, it activates the transforming growth factor-beta (*TGF-β*) signaling pathway, upregulates the expression of MMPs expression (redundant), and promotes ECM remodeling and myofiber regeneration.⁸⁶

It is worth further noting that the regulation of muscle repair by *SPP1* involves a broader mechanistic network. Studies have shown that *SPP1* upregulates MMP expression by activating the *TGF-β* signaling pathway, and MMP activity is finely regulated by endogenous tissue inhibitors of metalloproteinases (TIMPs), with the MMP/TIMP ratio determining the extent and direction of ECM remodeling.⁸⁸ During skeletal muscle regeneration, moderate ECM degradation facilitates myoblast migration to the site of injury, enabling the fusion and regeneration of new muscle cells.⁸⁹ Regular exercise increases the expression of *SPP1*, MMPs, and *TGF-β* in plasma, suggesting that exercise may promote muscle repair and regeneration through activation of the *SPP1-TGF-β-MMP* axis.⁹⁰

However, the *TGF-β* signaling pathway has a typical dual-edged sword characteristic. During aging, chronic persistent expression of *TGF-β* promotes tissue fibrosis and paracrine senescence, accelerating age-related functional decline.⁹¹ In fact, dysregulation of *TGF-β* protein and its related signaling components has been confirmed to be closely associated with chronic diseases, aging, and muscle atrophy under inactive conditions.⁹² In contrast, exercise-induced *TGF-β* activation is fundamentally different in nature: it is an acute, transient, and spatially confined physiological response aimed at initiating ECM repair and regenerative programs rather than driving pathological matrix deposition.⁹³ Exercise promotes the degradation of damaged ECM by upregulating MMPs while simultaneously activating *TGF-β* signaling to synthesize new ECM components.⁹⁴ This "degradation-synthesis" coupling is the core feature of adaptive remodeling.⁹⁵ Therefore, exercise-induced *TGF-β* regulation represents a functional, transient activation that is fundamentally distinct from the pathological chronic activation associated with aging.

It should be added that the regulation of muscle homeostasis by the *TGF-β* superfamily is highly complex.⁹⁵ This context-dependent functional difference depends on the duration (chronic vs. acute), intensity, and synergy with other signaling pathways (such as the *SPP1-MMP* axis).

Concurrently, exercise ameliorates age-related lipid metabolism disorders in muscle fibers, downregulates the expression of aging-related aberrant genes mediated by the *TNF-α* signaling pathway, and restores mitochondrial function in muscle fibers.⁶⁷ Aging is closely associated with the loss of muscle mass and strength, leading to decreased physical function and increased risk of frailty.⁹⁶ Resistance training has been proven to be effective in the elderly population for increasing muscle mass, enhancing strength, and improving physical performance.⁹⁷ In recent years, human studies have further revealed that exercise exerts significant time-dependent and fiber type-specific regulation on muscle fibers: a 24-hour time-series analysis found that the expression of protein-coding genes in human muscle fibers peaked at 8 h after resistance exercise, while ribosome-related genes remained enriched from 8 to 24 h post-exercise.⁹⁸ The transcription factor *MYC* mediates myofiber hypertrophy and optimization of metabolic phenotype by regulating ribosome biogenesis.^{98,99} These findings further refine the exercise-mediated myofiber regulatory network and provide a molecular basis for elucidating the differences in muscle adaptive responses under different exercise modalities. From an aging perspective, the temporal dynamics revealed by this 24-hour time-series analysis⁹⁸ provide important guidance for optimizing exercise intervention programs in the elderly population: understanding the peak time window of gene expression after resistance exercise can help design more

precise training frequencies and recovery periods to maximize muscle protein synthesis and hypertrophic adaptation; furthermore, the *MYC*-mediated ribosome biogenesis mechanism suggests a potential intervention target against aging-related ribosomal dysfunction, which is known to exist in aged muscle.⁹⁸

3.2.4. Functional regulation of MuSCs subsets

The regulation of *MuSCs* by exercise does not depend on alterations in overall cell number or cell cycle distribution, but rather alleviates aging-related *MuSCs* heterogeneity through targeted optimization of the composition and function of *MuSCs* subpopulations.⁹⁹ scRNA-seq studies have revealed that in aged skeletal muscle, the numbers of *MuSCs* subpopulations associated with glycolysis and mitochondrial respiration are significantly reduced, leading to metabolic phenotype imbalance and diminished regenerative potential of *MuSCs*.¹⁰⁰ In contrast, regular resistance training effectively restores the proportion of these two functional subpopulations and enhances the metabolic activity and activation capacity of *MuSCs* following injury by regulating the expression of immune response-related genes (e.g. *Il1r1* and *Cxcr4*).⁹⁹

This subpopulation-specific regulation also exhibits fiber type dependence: the analysis by Heidari et al.⁹⁹ confirmed that the activating effect of resistance training on *MuSCs* is more pronounced in Type II fast-twitch muscle fibers, enhancing the myonuclear fusion efficiency of *MuSCs* associated with fast-twitch fibers.⁹⁹ In contrast, endurance exercise preferentially regulates the function of *MuSCs* subpopulations associated with Type I slow-twitch muscle fibers.⁹⁹ This heterogeneous response spectrum precisely explains the controversies in traditional studies regarding the regulatory effects of exercise on different aging markers - a single marker cannot reflect the differential changes in *MuSCs* subpopulations, whereas single-cell technology can accurately capture such subpopulation-specific regulation, providing a molecular basis for developing personalized exercise intervention programs.

In summary, exercise-mediated regulation of aged skeletal muscle is a multidimensional, network-based, and precise intervention process: it is rooted in immune microenvironment remodeling, optimizes the *MuSCs* niche through restoration of intercellular communication and ECM remodeling, activates regenerative pathways by regulating myokine networks and the function of *MuSCs* subpopulations, and ultimately achieves comprehensive improvement in skeletal muscle metabolic capacity and regenerative ability. The application of single-cell and single-nucleus RNA sequencing technologies has not only revealed the cellular heterogeneity and molecular mechanisms masked by traditional mixed-sample studies but also established a complete regulatory cascade of "exercise → cellular response → molecular regulation". This provides strong support for in-depth understanding of the core mechanisms of exercise-mediated anti-aging effects and their translational medical applications.

Current studies utilizing single-cell and single-nucleus technologies to explore the effects of exercise on skeletal muscle aging still have certain limitations. First, existing studies (especially human studies) generally have small sample sizes. For example, the pioneering single-cell study by Lovric et al. included only three subjects, which limits statistical power and constrains the generalizability of its conclusions. Second, some studies did not explicitly report the specific post-exercise time point for biopsy collection, making it impossible to assess the temporal dynamics of transcriptional responses and limiting the precise understanding of the time window of acute exercise effects.⁶⁷ Third, the heterogeneity of subject populations, often uneven age composition (e.g. including both young and middle-aged/older individuals without stratified comparisons), and the frequent neglect of sex differences make it difficult to systematically assess the true differences in exercise effects across age and sex dimensions. Fourth, current studies are largely observational in design,⁶⁷ revealing primarily correlations between exercise and transcriptomic changes. For example, whether the observed changes in mesenchymal cells are a direct effect of exercise or a

secondary response to myofiber contraction or metabolic stress remains undetermined. These limitations suggest that future studies need to increase sample sizes, clearly report sampling time points, balance age and sex composition, and incorporate functional validation experiments (such as conditional gene knockout in animal models) to establish causal relationships.

4. Single-cell decoding of exercise-mediated central nervous system remodeling

The brain is the most complex organ in mammals, and recent large-scale single-cell studies have revealed that it contains hundreds of distinct cell states.¹⁰¹ Comparisons of brain cell populations across different ages have demonstrated how aging affects various brain cell types.¹⁰² Aging is not merely a functional decline of a single cell type, but rather a process characterized by synergistic dysfunction across multiple cell populations - including neurons, glial cells, immune cells, vascular-associated cells, and neural stem cells (NPCs) - ultimately leading to an imbalance in the brain cell ecosystem.^{103,104} The core manifestations of this imbalance include aberrant cell population sizes, phenotypic transitions, dysregulated molecular signaling, and impaired intercellular communication.¹⁰² The specific pathological alterations of each cell type can be precisely captured using single-cell omics technologies.^{105,106} Notably, aging-induced single-cell changes are not uniformly distributed throughout the central nervous system but exhibit significant regional heterogeneity: neuroblasts, glial cells, and immune cells in different brain regions display distinct vulnerabilities and temporal dynamics.¹⁰⁷ This section will thoroughly review the single-cell pathological features of aging across different brain regions, laying the foundation for subsequently elucidating the region-specific mechanisms of exercise intervention.

4.1. Single-cell atlas of the senescent nervous system

4.1.1. Single-cell atlas of the aging hippocampus

The hippocampus is a key hub for learning and memory and is one of the most vulnerable brain regions during aging.¹⁰⁸ Specifically, aging is associated with reduced hippocampal volume due to neuronal loss and decreased neurogenesis.¹⁰⁹ Single-cell studies have confirmed that the numbers of active neural stem cells (NSCs), neural progenitor cells (NPCs), and neuroblasts in the aged mouse brain are significantly reduced (by more than twofold).^{104,110}

Hippocampal neuroblasts exhibit early vulnerability: dentate gyrus neuroblasts markedly decrease within the first six months of mouse life, whereas olfactory bulb neuroblasts decline later in life.¹¹¹ This early decline suggests that the hippocampus is particularly susceptible to aging processes. The reduction in neurogenesis stems are not only from a decrease in NSCs numbers but also from impaired proliferation and differentiation capacity of NPCs.¹¹² At the molecular level, single-cell transcriptomic analysis has revealed that aged NPCs exhibit down-regulated expression of proliferation-associated genes (e.g. *Ei24*, *Arhgap11a*, *Cep57*) and neurogenesis-related pathway genes (e.g. *Prdm10*, *Nrg1*), a phenotype further exacerbated by dysregulated epigenetic regulation.¹¹³

Although the early reduction of dentate gyrus neuroblasts has been confirmed by multiple studies, these studies are primarily based on a single mouse strain.¹¹⁴ Whether mice of different genetic backgrounds exhibit the same aging trajectory remains unclear. Furthermore, a reduction in neuroblast number is not equivalent to a decline in neurogenic function - functional compensatory mechanisms (such as increased differentiation efficiency of remaining neuroblasts) may partially offset the negative impact of reduced numbers, but existing single-cell studies have paid insufficient attention to this possibility.

The aging phenotype of neurons themselves further exacerbates the homeostatic imbalance of the hippocampal microenvironment. With advancing age, the number of neurons in the brain gradually decreases,

and aged neurons exhibit typical pathological features, including enhanced p38 mitogen-activated protein kinase (MAPK) activity and altered growth factor expression profiles.¹¹⁵ At the molecular regulatory level, special AT-rich sequence-binding protein 1 deficiency is associated with p21-dependent senescence in dopaminergic neurons, while growth differentiation factor 11 deficiency may induce neuronal senescence by activating *CDKN1A* (encoding p21) transcription.^{116,117}

Chronically elevated insulin levels and amyloid- β (A β) accumulation are closely associated with the establishment of senescence-like phenotypes in neurons.¹¹⁸ Under normal conditions, insulin and insulin-like growth factor 1 inhibit glycogen synthase kinase 3 β through protein kinase B (Akt) phosphorylation, a process associated with preventing tau hyperphosphorylation; simultaneously, they reduce A β production by inhibiting β -site amyloid precursor protein cleaving enzyme 1 and its substrate amyloid precursor protein, and promote A β clearance.¹¹⁹ However, brain insulin resistance may disrupt this protective mechanism. Increased ceramide levels in neurons activate protein phosphatase 2A, inhibit Akt signaling, and block glucose transporter 4 translocation to the plasma membrane; furthermore, ceramide may also impede insulin binding to insulin receptor substrate 1 by activating MAPK and JNK phosphorylation, further exacerbating brain insulin resistance.¹²⁰ Therefore, increased ceramide content in neurons is closely associated with aggravated brain insulin resistance, A β deposition, and tau phosphorylation.¹²¹

A β deposition may participate in the neuronal senescence process through the following mechanisms: A β -induced microglial activation triggers the release of various inflammatory cytokines, which is a key process in neuroinflammation. Neuroinflammation is associated with synaptic dysfunction, inhibition of hippocampal neurogenesis, accelerated neuronal death, and the transport of A β and inflammatory mediators across the blood-brain barrier (BBB).¹²² The BBB is a dynamic barrier regulating material exchange between the periphery and the central nervous system and is crucial for maintaining brain homeostasis.¹²³ Following BBB disruption, hyperglycemia-induced peripheral oxidative stress triggers excessive ROS release from neuronal mitochondria, leading to oxidative stress.¹²⁴ Oxidative stress arises from an imbalance between ROS, reactive nitrogen species (RNS), antioxidants, and pro-oxidants, resulting in dysregulation of REDOX pathways and metabolic processes.¹²⁴ Prolonged oxidative stress is associated with exacerbated oxidative damage and senescence in neurons.¹²⁴ A β -induced oxidative stress can lead to further increases in ceramide levels (potentially forming a positive feedback loop), which is associated with hippocampal neuron senescence and even death.³⁰ Therefore, brain insulin resistance may participate in the establishment and consolidation of neuronal senescence phenotypes through a cascade of "ceramide positive feedback $\rightarrow$ A β /tau pathology $\rightarrow$ neuroinflammation $\rightarrow$ oxidative stress $\rightarrow$ mitochondrial dysfunction," and is associated with the onset and progression of Alzheimer's disease (AD).^{121,125}

In addition to progenitor cells, some previously understudied neuronal subpopulations also exhibit aging-related population changes, such as the newly emerging *Calcr1*⁺ neurons and *Smoc1*⁺ neurons in the aged brain.¹²⁶ Notably, *Smoc1*, as a specific marker, is highly enriched in the cortical tissue of human AD patients and co-localizes with A β plaques, suggesting that this novel neuronal subpopulation may be involved in the pathogenesis of age-related neurodegenerative diseases.¹²⁷

Regarding the immune microenvironment, recent single-cell studies have revealed the expansion of T cells in the hippocampal neurogenic niche, potentially due to BBB impairment during aging.¹²⁶ These T cells express IFN- γ , which is associated with reduced NSC proliferation.¹¹⁶ Furthermore, natural killer (NK) cells accumulate (twofold or more) in the aged human and mouse hippocampal dentate gyrus, where they participate in the clearance of neuroblasts and are associated with cognitive decline during brain aging.¹⁰⁴

Although the accumulation of T cells and NK cells in the hippocampus is associated with cognitive decline, single-cell technologies

cannot distinguish whether these immune cells actively infiltrate or passively enter (secondary to BBB injury).¹¹⁶ Moreover, there is a lack of functional experiments demonstrating whether depletion of these immune cells can reverse the decline in neurogenesis - a key piece of evidence for establishing causality. Future studies combining single-cell technologies with *in vivo* immune interventions (e.g. CD8⁺ antibody-mediated T cell depletion) will help address this question.¹²⁸

The discovery of disease-associated astrocytes (DAAs) and disease-associated microglia (DAMs) represents a major breakthrough of single-cell technologies.¹¹⁶ However, whether these cell states represent novel cell subpopulations that emerge during aging or are functional responses of normal cells to microenvironmental changes (e.g. inflammation-induced activation states) remains an open question. Single-cell transcriptomic data reflect transient transcriptional states rather than stable cellular identities. Spatial transcriptomics and lineage tracing studies will help distinguish between these possibilities.¹²⁹

4.1.2. Single-cell atlas of the aging subventricular zone (SVZ) and white matter

The SVZ is another key neurogenic niche in the adult brain, and its adjacent white matter regions also exhibit significant cellular dynamics during aging.¹³⁰ The reduction of SVZ neuroblasts shows a temporally delayed pattern: unlike the hippocampal dentate gyrus, olfactory bulb neuroblasts (originating from the SVZ) decline later in life.¹³⁰ In addition to changes in NSCs and NPCs, the oligodendrocyte lineage exhibits high susceptibility to aging in the SVZ and white matter regions, considered a core cause of white matter dysfunction in the aged brain.¹³⁰ Aging significantly affects oligodendrogenesis, reducing the proliferation rate of oligodendrocyte precursor cells (OPCs) and their differentiation into functional cell types, particularly in white matter and ventricular regions.¹³¹ Meanwhile, in the aged mouse brain, cycling OPCs (*Mki67*⁺), committed OPCs (*Bmp4*⁺*Enpp6*⁺), and newly formed OPCs (*Prom1*⁺*Tcf7l1*⁺) are significantly reduced (twofold or more).^{109,117}

The discovery of disease-associated oligodendrocytes (DAOs) has sparked debate regarding their functional significance: Are DAOs an adaptive response during aging (e.g. attempting to repair myelin damage) or purely pathological alterations? Current single-cell data cannot answer this question. Furthermore, whether the upregulation of DAO markers (e.g. *Serpina3n*) during aging and in AD is cell-autonomous or induced by inflammatory factors secreted by adjacent microglia remains to be validated through lineage-specific gene knockout experiments.

At the immune cell level, the aged brain is characterized by the expansion and activation of various immune cells, particularly prominent in the SVZ and white matter regions.¹¹⁶ Aged microglia exhibit increased proliferation rates, transitioning to an APOE⁺CSF1⁺ subtype with enhanced expression of genes associated with immune activation and phagocytosis.¹¹⁶ This subtype, termed DAMs, increases significantly (more than fourfold) in the aged mouse brain, particularly in white matter, and is frequently found near A β plaques in the brains of mice and human AD patients.¹¹⁶ In addition to DAMs, a notable feature of the aged mouse brain is the expansion of lymphocyte-dependent interferon-responsive microglia near CD8⁺ T cells in aged white matter.¹¹⁶ The expansion of T cells in neurogenic niches (including the SVZ region) is also associated with BBB impairment.¹²⁴ These T cells express interferon- γ , which is associated with reduced NSC proliferation.¹²⁴ The expansion of immune cells - including microglia, macrophages, T cells, and NK cells - may have dual effects. While they benefit brain health by clearing debris and A β plaques, their persistent activation may lead to chronic inflammation, impaired tissue repair, and increased risk of neurodegenerative diseases.^{125,132}

Notably, most single-cell immune profiling studies have employed whole-brain tissue dissociation methods, which may result in the loss of spatial information regarding meningeal, perivascular, and parenchymal immune cells.¹³³ Immune cells from different anatomical locations may have entirely distinct functions and aging dynamics. For

example, the aging phenotypes of meningeal macrophages and parenchymal microglia may differ significantly, but conventional single-cell sequencing cannot distinguish between them.¹³³ The development of spatial transcriptomics and *in situ* sequencing technologies will offer possibilities to address this issue.

4.1.3. Single-cell characteristics of other aging brain regions and structures

Beyond the hippocampus and SVZ, aging significantly affects other brain regions and structures, some of which show high pathological overlap with neurodegenerative diseases.

In the pituitary gland, recent single-cell studies have found a reduction (more than twofold) in pituitary cell numbers in the aged mouse brain, and a decreased proportion of growth hormone-secreting cells (somatotrophs) in the aged human pituitary, accompanied by pituitary atrophy and reduced hormone secretion.¹⁰⁹ Pituitary hormones are critical for regulating brain homeostasis and systemic aging; their insufficient secretion may trigger metabolic disorders, sarcopenia, and cognitive decline, further exacerbating aging-related neuropathological changes.¹³⁴

Compared to other brain regions, single-cell studies on cortical and pituitary aging are relatively limited, with small sample sizes and potentially insufficient statistical power. In summary, aging-induced single-cell alterations in the central nervous system exhibit significant regional heterogeneity. These single-cell pathological features suggest that cellular ecosystem imbalance in the aged nervous system is a core pathological basis for the onset and progression of neurodegenerative diseases. In the hippocampus, early neuroblast reduction, T cell and NK cell infiltration, increased DAAs, and insulin/A β -related neuronal senescence, representing a core region for aging-related neurogenic decline and cognitive impairment.¹¹¹ These pathological features are also observed in the SVZ and adjacent white matter. Oligodendrocyte lineage dysregulation (reduced OPCs, increased DAOs), expansion of DAMs and CD8⁺ T cells, manifesting as loss of white matter integrity and an exacerbated inflammatory microenvironment.¹³⁵ In the cerebral cortex, emergence of Smoc1⁺ aging-associated neurons co-localizing with A β plaques.^{136,137} The aging phenotypes of neurons, astrocytes, OPCs, and microglia are interconnected and may act synergistically through multiple mechanisms. Widespread release of SASP factors in the brain may create a chronic inflammatory microenvironment associated with NSC dysfunction, neuronal damage, and glial cell phenotypic transitions, while vascular dysfunction may further exacerbate insufficient nutrient supply and accumulation of pathological factors.¹³⁵ These changes are associated with the progression of neurodegenerative diseases such as AD, Parkinson's Disease (PD), and Amyotrophic Lateral Sclerosis (ALS) (Fig. 2).¹³⁷

Among these regions, the hippocampus (especially the dentate gyrus) and SVZ, as the two major neurogenic niches in the adult brain, are sensitive to aging, and have been shown to respond positively to exercise intervention.¹³⁸ Neurogenic decline in these two regions occurs early and is severe, and their immune microenvironment dysregulation and OPCs dysfunction can be modulated by exercise.¹³⁸ Therefore, the next section will focus on elucidating the mechanisms by which exercise remodels the cellular ecosystem of the hippocampus and SVZ, revealing the protective effects of exercise on the central nervous system at the single-cell level.¹³⁹

Although single-cell omics technologies have revealed the cellular heterogeneity of the aged brain at unprecedented resolution, and regional heterogeneity has become a consensus in aging neuroscience, current studies have several important limitations that warrant careful interpretation. First, there are certain technical limitations. Single-cell transcriptomic analysis relies on tissue dissociation processes that may introduce cell type-specific biases - certain fragile cell types (e.g. mature neurons) may be lost or undergo transcriptional alterations during dissociation.¹³³ Although snRNA-seq avoids tissue dissociation bias, its captured transcripts originate primarily from the nucleus and may not fully reflect dynamic changes in the cytoplasm (e.g. local synaptic

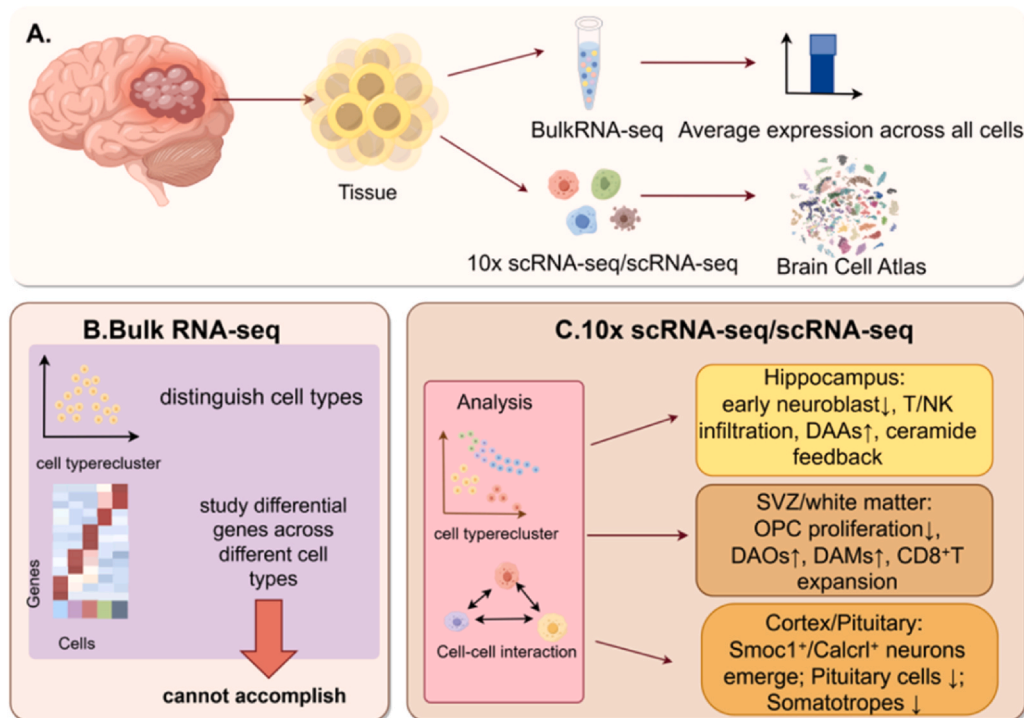


Fig. 2. Overview of transcriptomic strategy comparison and region-specific aging-associated cellular remodeling in the brain. (A) Experimental sequencing workflow: Tissue samples collected from distinct brain regions were subjected to two transcriptomic profiling approaches. Conventional bulk RNA sequencing (Bulk RNA-seq) can only detect averaged gene expression across the entire tissue, whereas 10 × scRNA-seq/snRNA-seq dissociates samples at single-cell resolution to construct a high-precision, comprehensive brain cell atlas. (B) Technical limitations of Bulk RNA-seq: Bulk transcriptome analysis cannot distinguish the highly complex cellular subtypes within brain tissue, fails to accurately resolve cell-type-specific differential gene expression signatures, and is therefore unable to decipher critical biological changes driven by cellular heterogeneity. (C) Technical strengths and key discoveries of 10 × single-cell transcriptome sequencing: Single-cell sequencing enables systematic cell subtype clustering, differential gene expression analysis, and cell-cell interaction network reconstruction, allowing precise identification of region-specific aging-related cellular alterations across brain compartments. Specifically, the hippocampus exhibits decreased early neuroblast numbers, accompanied by T/NK cell infiltration, elevated proportion of disease-associated astrocytes (DAAs), and abnormal activation of the ceramide feedback loop; in the SVZ and white matter, OPCs proliferation is diminished, DAOs and DAMs are markedly increased, alongside prominent CD8⁺ T cell expansion; in the cerebral cortex and pituitary gland, a unique *Smoc1*⁺/*Calcr1*⁺ neuronal subpopulation specifically emerges, while pituitary cells and somatotropes are significantly reduced. (By Figdraw).

translation).¹³³ Second, sample homogeneity and the challenges of cross-species comparison remain important considerations. Most current single-cell studies focus on mice or single regions of the nervous system, lacking systematic analysis of complete human brain tissue.¹³³ Mice and humans exhibit significant differences in cell type markers, transcriptional regulatory networks, and aging dynamics. For example, whether DAAs/DAMs markers defined in mice have the same cellular identity and functional significance in humans requires systematic validation. Furthermore, most existing studies employ mice aged 12–18 months, leaving it unclear whether these findings apply to very old aged populations (over 24 months). The construction of single-cell atlases of complete human brain tissue and cross-species comparative studies are important future directions. Third, there is a notable lack of functional validation. Single-cell studies primarily provide correlational evidence; the functional significance of newly discovered cell subpopulations requires validation through experimental approaches such as gene knockout/overexpression and chemogenetics. Therefore, the cell type changes described in this section should be understood as correlational findings, and their causal relationships await confirmation through functional experiments.

4.2. Single-cell omics reveals mechanisms of exercise intervention in the aged brain

As a well-validated intervention, exercise can effectively restore the function of multiple tissues, delay or reverse age-related functional decline. Its protective effects on the brain and the underlying molecular mechanisms have become a research hotspot.^{140–142} In recent years, the

application of scRNA-seq, snRNA-seq, and spatial transcriptomics technologies has provided a precise perspective for dissecting the impacts of exercise on brain cell composition, gene expression, and molecular regulatory networks.

4.2.1. Mechanism of cellular remodeling and molecular regulation of the aging hippocampus by exercise

As a key center for learning and memory, the functional homeostasis of the hippocampus is highly dependent on the coordinated regulation of neurogenesis and synaptic plasticity. For example, a single-nucleus study in male mice found that after four weeks of voluntary wheel running exercise, the cellular composition and transcriptomic profile of the hippocampus in 3-month-old and 18-month-old mice were significantly altered.¹⁴³

At the cellular compositional level, exercise specifically modulates the proportions of hippocampal neuron subtypes, as the proportion of inhibitory interneurons (InN2) decreases, while that of excitatory neurons (ExN11) with a newborn/immature phenotype increases markedly.¹⁴³ This shift is closely linked to the regulation of the transcription factor *Prdm16*, as the population of InN2 neurons with high *Prdm16* expression in sedentary mice is significantly reduced after exercise.¹⁴³ Previous studies have suggested that high *Prdm16* expression may characterize a state of “developmental pause” during neurogenesis, and exercise may relieve this blockade by downregulating *Prdm16*, accelerating the maturation of newborn neurons, thereby increasing the proportion of ExN11 and enhancing hippocampal neurogenesis.¹⁴⁴

At the functional level of mature neurons, exercise also exerts critical regulatory effects. Excitatory neurons in the hippocampal CA

region exhibit widespread transcriptomic changes following exercise, among which *Hes1*, a key effector of the notch signaling pathway, is significantly upregulated.¹⁴⁵ The notch pathway is an important signaling axis governing synaptic plasticity; this finding suggests that exercise may enhance the synaptic structure and function of mature neurons by activating this pathway, thereby optimizing the information-processing efficiency of hippocampal neural networks.¹⁴⁵ The synergy of these two pathways collectively forms the molecular basis for exercise-mediated improvements in hippocampal function and cognitive ability.

Furthermore, the potential rejuvenating effect of exercise on the aging hippocampus is also reflected in the regulation of the immune microenvironment. Another single-cell transcriptomic study focused on female C57BL/6 J mice, comparing the cellular responses of the hippocampus between 3-month-old (young adult) and 18-month-old (aged) mice.¹⁴⁵ This study employed an optimized exercise intervention protocol: aged running mice were allowed free access to running wheels for 21 consecutive days, followed by a 14-day rest period (a duration designed to allow exercise-induced neural precursor cells to differentiate into immature hippocampal neurons), after which tissue collection was performed.⁶⁷ Through single-cell transcriptomic sequencing of three samples per group combined with immunohistochemical validation, the study found that the number of T cells in the hippocampus of aged running mice was significantly reduced, and exercise had the most profound transcriptional impact on microglia.⁶⁷ It is important to emphasize that these findings are based on a female animal model, and their extrapolation to male individuals or human physiological/pathological states requires further validation. Notably, exercise had the most significant transcriptional effect on microglia, and the reduction in infiltrating T cells may be one of the key factors mediating the transcriptional reprogramming of microglia.¹⁴⁶

4.2.2. Cellular remodeling and molecular regulation mechanisms of exercise in the aging SVZ

The remodeling effects of exercise on the brain are not limited to the hippocampus, but extend to multiple brain regions and the spinal cord, exerting a more pronounced impact on the central nervous system of aged individuals.⁶⁷ As a critical neurogenic niche in the adult brain, SVZ relies on the functional homeostasis of its niche cells to regulate the activity of NSCs.⁶⁷ Single-cell omics studies have confirmed that exercise restores the cellular function and proportional equilibrium of the SVZ niche via transcriptomic regulation, while alleviating the local inflammatory microenvironment - thereby establishing a dual regulatory effect of “proliferative activation + inflammatory inhibition”.^{67,146} A study subjected young (3-month-old) and aged (19-month-old) male C57BL/6 mice to a 45-day voluntary wheel running exercise intervention (a relatively chronic long-term exercise training paradigm).⁶⁷ Post-intervention scRNA-seq analysis revealed that the proportions of multiple cell types in the SVZ niche of aged exercised mice exhibited a trend toward restoration to a young sedentary state, and the proportions of activated NSCs and neuroblasts were significantly higher in the exercise groups (both young and aged) compared to their corresponding sedentary controls.^{67,147} This finding is consistent with previous studies reporting that exercise promotes neurogenesis.^{138,148}

Another study focusing on the SVZ further explored the potential mechanisms by which exercise improves NSC function.¹⁴⁷ This study, conducted in male mice, compared the SVZ responses of young (4-month-old) and aged (22-month-old) individuals after five weeks of voluntary wheel running.¹⁴⁷ The results showed that exercise was associated with downregulated expression of inflammation-related genes in NSCs and niche cells within the SVZ of aged mice, accompanied by inhibition of age-associated amplification of neuroinflammation.¹⁴⁷ At the molecular mechanistic level, it is speculated that exercise may attenuate the toxic effects of pro-inflammatory factors on NSCs by modulating the interactions between immune cells and NSCs, thereby partially restoring their regenerative activity.¹⁴⁷ Integrated with correlational studies, it is hypothesized that systemic factors mediate

this inflammatory remodeling effect. Exercise elevates blood levels of complement cascade inhibitors, GPLD1, and other molecules. After crossing the blood-brain barrier, these molecules may reduce the proportion of pro-inflammatory microglia. This process could create a stable microenvironment for SVZ neural stem cells and ultimately support neurogenic capacity and cognitive function in aged individuals.⁶⁷

In summary, exercise remodels and regulates the aged hippocampus and SVZ through a multidimensional, networked, and precise intervention process. This process centers on optimizing cell subtype proportions, activating molecular pathways, and remodeling the immune microenvironment. By regulating cell-cell interactions and systemic factor-mediated signaling, exercise synergistically improves the functional homeostasis of these two key neurogenic brain regions. Ultimately, this provides support for the recovery of neurogenic capacity and the maintenance of cognitive function in the aged brain. The application of single-cell omics technologies clearly delineates the details of cellular heterogeneity and molecular regulation across distinct brain regions, and establishes a complete regulatory cascade of “exercise → cell subtype response → molecular pathway activation → microenvironment optimization”. This lays a solid foundation for in-depth elucidation of the core mechanisms underlying exercise-mediated brain aging delay and associated clinical translational research.

While acknowledging the above progress, it is essential to recognize several key limitations in the current field that need to be systematically addressed in future research. At the technical level, the tissue dissociation process of scRNA-seq introduces cell type selection bias, potentially leading to the loss of fragile neurons or enrichment of dissociation-resistant cells; although snRNA-seq avoids this issue, it results in the loss of cytoplasmic and synaptic transcripts. Furthermore, most studies sequence brain regions as whole entities, neglecting the spatial heterogeneity of hippocampal subregions and different microanatomical locations of the SVZ. At the experimental design level, exercise intervention protocols vary significantly across studies (duration ranging from 4 weeks to 45 days, with or without recovery periods, and varying lengths of recovery periods), severely limiting cross-study integration and direct comparison. It is currently difficult to distinguish whether observed effects are direct consequences of exercise itself or indirect manifestations of developmental / differentiation processes enabled by the post-exercise recovery period. Moreover, there is a lack of unified standards for defining “aged” (ranging from 18 to 22 months, corresponding to different aging stages in mice), and most studies focus on male animals, exhibiting significant sex bias. At the causal validation level, the vast majority of current findings are based on correlational analysis. Although single-cell transcriptomics has revealed significant associations between exercise and specific molecular and cellular changes, direct causal experiments (e.g. cell type-specific gene knockout, chemogenetic blockade, or pharmacological intervention) are lacking to validate whether these changes are necessary for the functional benefits of exercise. Future research should focus on breakthroughs in the following directions: (1) integrating single-cell transcriptomics with spatial transcriptomics to construct exercise response maps at the sub-anatomical level of brain regions; (2) systematically comparing the differential effects of various exercise doses (type, intensity, duration, frequency) as well as sex and age stratification; (3) combining single-cell atlas analysis with functional causal validation experiments to establish a complete causal chain from molecular events to behavioral improvements. These efforts will lay a more solid foundation for in-depth elucidation of the core mechanisms underlying exercise-mediated brain aging delay and associated clinical translational research.

5. Single-cell omics decodes exercise-mediated vascular remodeling

5.1. Single-cell atlas of the aged cardiovascular system

The primary function of the cardiovascular system is to maintain blood circulation and supply oxygen and nutrients throughout the

body.¹⁴⁹ Cardiovascular cells mainly include endothelial cells and SMCs (also known as mural cells), in addition to cardiomyocytes, pericytes, fibroblasts, and immune cells in the heart.¹⁹ Although their aging mechanisms involve common features such as activation of inflammatory pathways, dysregulated intercellular communication, and impaired protein homeostasis, they exhibit highly heterogeneous phenotypes across different anatomical regions.³⁵

In the SVZ, a neurovascular microenvironment, single-cell transcriptomic studies have shown that aged endothelial cells exhibit transcriptional upregulation of *TNF α* , *IFN γ* , and hypoxia-related genes, changes that are associated with the formation of a chronic inflammatory microenvironment.¹⁴⁰ Concurrently, the expression of the TNF-like weak inducer of apoptosis (TWEAK) signaling ligand (TNFSF12) and its receptor *TNFRSF12A* is downregulated, suggesting that their signaling crosstalk with surrounding cells such as NSCs may be weakened, thereby affecting vascular homeostasis.^{150,151} Furthermore, the expression of heat shock protein 90 family genes is consistently downregulated, implying an impaired ability to maintain protein folding and stability.¹⁴⁰ Aged SMCs in this region primarily exhibit decreased activity of the ECM signaling pathway, which may affect the structure and function of the perivascular microenvironment.¹⁵² It should be noted that these transcriptomic changes mainly reflect correlation rather than causation, and it remains unclear whether these alterations represent the emergence of unique aged cell subsets or merely functional state transitions of existing cell subsets.

In the skeletal muscle vascular bed, aged SMCs display a distinct transcriptional signature – abnormally enhanced expression of genes associated with communication with immune cells and stem cells via the CXCR chemokine receptor (CXCR) signaling pathway, which is speculated to potentially disrupt local microenvironmental balance.¹⁵³ However, these findings are primarily derived from mouse models, and their conservation in humans, as well as whether these gene expression changes directly lead to functional alterations, remain to be validated.

In the heart, aging involves not only vascular cells but also coordinated changes across multiple cell types.¹⁵³ Single-cell transcriptomic studies have revealed cellular and histological features of cardiac aging in non-human primates (cynomolgus monkeys): compared with young (4–6 years old) monkeys, aged (18–21 years old) hearts exhibit myocardial fatty infiltration, reduced cardiomyocyte number, increased fibroblast proportion, and enhanced immune cell infiltration, changes associated with myocardial fibrosis.^{154–156} Histological detection further showed significantly elevated levels of SA- β -Gal staining in aged hearts, suggesting accumulation of senescent cells.³⁷ Although these primate data are closer to humans than rodent models, the limited sample size (only 2–3 animals per group) warrants further exploration of their statistical power and generalizability.

In the systemic macrovasculature and cerebrovascular system, features of vascular aging include impaired aortic endothelial function, reduced vascular cells in the brain, and compromised BBB function.¹⁵³ Single-cell and spatial transcriptomic studies have further revealed transcriptional changes in large vessel cells in the context of atherosclerosis.¹⁵³ In a mouse model of atherosclerosis, researchers identified four cell populations enriched for aging-related gene signatures, including vascular SMCs, fibroblasts, and T cells, and constructed an atherosclerosis-associated gene set (e.g. *Spp1*, *Ctsb*, *Tnfrsf11b* mRNAs).¹⁵⁵ These cells were enriched for pathways related to TGF β signaling, ECM-receptor interaction, and epithelial-mesenchymal transition, which are speculated to participate in pathological vascular remodeling through ECM remodeling.¹³ Spatial transcriptomics validated the enrichment of these mRNAs in plaques.¹⁵³ Notably, senolytic clearance of senescent cells significantly reduced vascular stiffness, providing experimental support for a causal relationship between senescent cells and vascular function.¹⁵⁷ However, these transcriptomic findings remain largely based on correlational analyses. Currently, there is an ongoing debate about whether the observed gene expression changes represent unique senescent cell subsets or instead reflect functional state transitions of existing cells.

In summary, aging of the cardiovascular system is a complex process involving multiple regions, multiple cell types, and distinct but interconnected mechanisms. Single-cell technologies provide unprecedented resolution for depicting this process.

5.2. Exercise-mediated cellular and molecular regulatory mechanisms in the aged cardiovascular system

Exercise exerts multifaceted regulatory effects on the aging of cardiovascular cells, with its restorative effects also exhibiting region- and cell type-specificity.¹⁵⁸ It should be noted that the following findings are primarily derived from single-cell transcriptomic studies in mouse models, and their translation to humans requires caution; the described changes mainly reflect correlations, and causal relationships await functional experimental validation.

In the SVZ: data from a mouse single-cell transcriptomic study show that four weeks of sustained exercise upregulates the expression of the anti-inflammatory cytokine C-X3-C motif chemokine ligand 1 (CX3CL1), attenuates IFN-induced neurotoxicity by activating the CX3C signaling pathway, and indirectly protects vascular-associated neuroregulatory networks.¹¹³ For SMCs, although exercise does not appear to directly reverse the age-related decline in ECM signaling activity, it may indirectly enhance their function by improving the global intercellular communication network.¹⁵² It remains unclear whether these transcriptional changes represent the emergence of unique cell subsets or functional state transitions of existing cell subsets.

In the skeletal muscle vascular bed, exercise is associated with the remodeling of intercellular communication of SMCs, potentially restoring their normal interactions with immune cells and stem cells by regulating the CXCR signaling pathway.¹ Concurrently, exercise-mediated secretion of SPP1 by muscle fibers can act on surface receptors on SMCs, regulating the expression of their metabolism- and immune-related genes and participating in the repair of vascular homeostasis.¹⁵⁹ These findings are primarily derived from mouse models, and their conservation in humans awaits validation.

In the heart and systemic vasculature, exercise is also associated with a reduction in multi-tissue inflammation.¹ A single-cell transcriptomic study subjected young (4-month-old) and aged (22-month-old) male C57BL/6J mice to 5 weeks of voluntary wheel running intervention and found that exercise significantly reduced the activity of inflammatory pathways in hematopoietic stem/progenitor cells of aged mice and decreased immune cell infiltration in vascular-related brain regions, thereby improving the microenvironment.^{67,160} The protective effects of exercise on aortic endothelial function and maintenance of BBB function together constitute the molecular basis for its attenuation of systemic vascular aging.¹⁵⁸ However, the differences between acute and chronic exercise remain unclear. Moreover, most experiments have used male mice, and the impact of sex differences on exercise effects requires systematic evaluation.

The regulatory effect of exercise on aging has certain limitations, particularly in the hematopoietic system, characterized by "partial improvement with functional resistance." Single-cell transcriptomic studies have confirmed that exercise significantly reduces the activation of inflammatory pathways in aged mouse hematopoietic stem/progenitor cells and immune cells and shifts the transcriptome toward a younger state.⁶⁷ However, these studies also found that exercise failed to fundamentally correct core functional defects of aged hematopoietic stem cells (HSCs) – although exercise altered the composition of HSCs downstream progenitors (shift toward MPPs), HSCs frequency itself did not change significantly, and exercise-induced improvement in HSCs self-renewal capacity and myeloid differentiation bias remained incomplete.⁶⁷

This phenomenon suggests that the senescent phenotype of aged HSCs is controlled by more complex mechanisms beyond inflammatory regulation. Based on the available literature, we speculate that these mechanisms may involve the following aspects. First, accumulation of

oxidative stress: aging leads to elevated levels of reactive oxygen species within HSCs, causing persistent DNA damage and telomere attrition, impairing self-renewal capacity. Although exercise can partially improve the antioxidant defense system, accumulated oxidative damage may not be fully cleared. Second, intrinsic signaling deficits: dysregulated activity of signaling pathways such as Wnt/ β -catenin, p38 MAPK, and mTORC1 in aged HSCs, along with altered expression of epigenetic regulators, drive myeloid-biased differentiation and loss of quiescence. These intrinsic programmatic changes may be relatively insensitive to exercise intervention. Third, non-immune paracrine signals: paracrine factors produced by stromal cells (e.g. osteoblasts, mesenchymal stem cells) in the aged bone marrow microenvironment are abnormally altered, and these microenvironmental changes may regulate HSCs function beyond the effects of exercise-mediated immune modulation. Notably, although single-cell transcriptomic studies have confirmed the profound impact of exercise on circulating and tissue-resident immune cells and have shown that exercise improves the up-regulation of multiple inflammatory pathways and restores immune cell-mediated intercellular communication in aged animals,⁶⁷ these studies did not examine changes in bone marrow stromal cells, which may be one reason why exercise fails to fully reverse the aged HSCs phenotype.

In summary, aging of the cardiovascular system is a complex process involving multiple regions, multiple cell types, and distinct but interconnected mechanisms. Exercise achieves partial functional remodeling of senescent cells by targeting key signaling pathways and reconstructing intercellular communication networks. However, the phenomenon of "functional resistance" observed in the hematopoietic system reminds us that the benefits of exercise intervention have boundaries. In-depth understanding of the above resistance mechanisms – particularly the irreversibility of oxidative damage, the aging of intrinsic cellular programs, and the persistence of non-immune microenvironmental signals – will provide a theoretical foundation for developing "exercise + " combination intervention strategies (e.g. exercise combined with senolytics, antioxidants, or epigenetic modulators).

6. A single-cell perspective on the systemic effects of exercise on multiple organs/tissues and intercellular and inter-organ communication

Based on the tissue-specific responses described above, this section integrates existing single-cell transcriptomic datasets to systematically elucidate how exercise exerts synergistic anti-inflammatory and anti-aging effects across multiple tissues via intercellular and inter-organ communication networks. It should be noted that the exercise interventions in the cited studies are primarily aerobic exercise modalities (voluntary wheel running or treadmill exercise), but they differ in intervention duration, training period, and animal age. These differences provide an important reference for understanding the relationship between exercise dose and biological effects.

One of the most core systemic effects of exercise on the aging organism is the repair of damaged intercellular communication networks. This repair occurs simultaneously across the skeletal muscle, nervous, and hematopoietic systems, exhibiting highly similar molecular patterns.⁶⁷ At the skeletal muscle level, following a 5-week voluntary wheel-running exercise intervention in 22-month-old aged male mice (with activity during the dark phase and running distance monitored by a digital odometer), single-cell transcriptomic analysis showed that exercise significantly enhanced the efficiency of ligand-receptor interactions between macrophages and MuSCs, restored the paracrine signaling activity of key pathways such as notch and TGF- β , and repaired the signaling capacity of fibro-adipogenic progenitors, Schwann cells, and endothelial cells.⁶⁷ Notably, this communication repair is not an isolated event in skeletal muscle. In the nervous system, after 45 days (in 19-month-old mice) or 5 weeks (in 22-month-old mice) of voluntary wheel running, the interaction between microglia and NSCs was

similarly restored, and the toxic effects of pro-inflammatory factors on NSCs were significantly reduced.¹⁴⁴ In the hematopoietic system, the same exercise intervention period remodeled the communication between stromal cells and HSCs within the bone marrow immune niche, restoring the balance between lymphoid and myeloid differentiation.⁶⁷

At the vascular level, the communication-repairing effect of exercise is also significant. Five weeks of voluntary wheel running restored normal interactions between SMCs, immune cells, and stem cells in the skeletal muscle vascular bed.⁶⁷ This effect is partly mediated by exercise-induced myokine SPP1, which binds to receptors on the surface of SMCs, regulating the expression of metabolic and immune-related genes and participating in the repair of vascular homeostasis.⁶⁷

As a physical platform for intercellular communication, the ECM undergoes rapid remodeling in response to exercise stimulation. An acute exercise study in humans showed that a single session of 3 $\times$ 30-second Wingate sprints (with 2-minute rest intervals) induced a strong transcriptional response in mesenchymal cells within skeletal muscle, with the differentially expressed genes significantly enriched in ECM remodeling pathways.⁸⁰ This rapid response suggests that ECM remodeling is one of the early adaptive reactions to exercise. Following 5 weeks of voluntary wheel running, this remodeling effect is further consolidated.⁸⁰ Exercise drives the degradation of damaged ECM components by upregulating the expression of MMPs and promotes the synthesis of new ECM proteins, thereby providing structural support for intercellular signal transduction.⁶⁷ The study by Liu et al.⁶⁷ confirmed that ECM remodeling also occurs in the NSCs niche.

Aging induces a highly consistent inflammatory signature across multiple tissues, characterized by the systemic upregulation of pro-inflammatory pathways such as Nuclear Factor-kappa B (NF- κ B) and TNF- α .⁶⁷ Exercise can reverse this signature at single-cell resolution, and this reversal exhibits tissue-specific and cell-type-specific heterogeneity. In skeletal muscle, following 5 weeks of voluntary wheel running in 22-month-old aged mice, single-cell analysis showed that exercise significantly reduced the infiltration of pro-inflammatory macrophages (CD11b⁺Ly6C⁺) and neutrophils, increased the proportion of CD206⁺ anti-inflammatory macrophages (M2 phenotype), and restored the physiological number of muscle-resident CD206⁺ monocytes.¹⁴⁷ This inflammatory remodeling effect is also present in the central nervous system. In 19-month-old aged mice after 45 days of voluntary wheel running, the number of T cells in the hippocampus was significantly reduced, the transcriptome of microglia was profoundly altered, and the proportion of pro-inflammatory microglia decreased, creating a stable microenvironment for NSCs.¹⁴⁷ In the hematopoietic system, 5 weeks of voluntary wheel running reduced the activity of inflammatory pathways in aged mouse HSCs and reversed the age-related myeloid bias, thereby reducing the production of pro-inflammatory immune cells at the source.⁶⁷

The heterogeneity of cellular metabolism is a long-neglected dimension of the aging process. Single-cell data reveal that different cell subpopulations within the same tissue exhibit distinct metabolic phenotypes, and exercise can selectively regulate the proportion and function of these metabolic subpopulations.⁹⁸ At the skeletal MuSCs level, single-cell analysis of aged skeletal muscle revealed a significant reduction in the numbers of MuSCs subpopulations associated with glycolysis and mitochondrial respiration, leading to metabolic imbalance and diminished regenerative potential.⁹⁸ Regular resistance training (human studies show that gene expression in MFs peaks 8 h after a single resistance exercise session, with ribosome-related genes being enriched between 8 and 24 h post-exercise) can effectively restore the proportion of these two functional subpopulations and enhance the metabolic activity of MuSCs by regulating the expression of immune response-related genes such as *Il1r1* and *Cxcr4*.⁹⁹ This regulation is fiber-type dependent: resistance training preferentially activates stem cell subpopulations associated with Type II fast-twitch fibers, while endurance exercise preferentially regulates the function of stem cell subpopulations associated with Type I slow-twitch fibers.¹⁴⁷ This

finding provides a single-cell-level explanation for the differential effects of different exercise modalities.

Similar selective regulation of metabolic subpopulations exists in NSCs and HSCs. In aged mice subjected to 5 or 45 weeks of voluntary wheel running, the proportion of activated NSCs and neuroblasts in the SVZ niche was significantly increased, a recovery associated with the reprogramming of metabolism-related gene expression.¹⁴⁷ In the hematopoietic system, exercise reverses the age-related myeloid bias, a process that essentially involves a rebalancing of HSCs metabolic preference from glycolysis towards oxidative phosphorylation.⁶¹ At the parenchymal cell level, the metabolic regulatory effects of exercise are also significant. Five weeks of voluntary wheel running improved age-related lipid metabolism disorders and restored mitochondrial function in aged mouse MFs.¹⁴⁷ Human resistance exercise studies further revealed that the expression of protein-coding genes in MFs peaks 8 h after exercise, while ribosome-related genes are continuously enriched from 8 to 24 h.^{98,147} The transcription factor *MYC* mediates myofiber hypertrophy and optimization of the metabolic phenotype by regulating ribosome biogenesis. In cardiomyocytes and vascular endothelial cells, the 12-month long-term exercise study by Sun et al.¹⁴⁷ showed that exercise resets the rhythmic metabolic network by restoring *BMAL1* expression, including the circadian rhythms of glucose metabolism, lipid metabolism, and mitochondrial bioenergetics.

Age is a key variable determining the metabolic response pattern. The study by Sun et al.¹⁴⁷ found that young mice exhibit a more pronounced metabolic response to exercise, with metabolic tissues such as the aorta, kidney, small intestine, and testis showing the most robust transcriptional changes in young individuals. In contrast, aged individuals benefit most from the protection of the central nervous system, and their metabolic response is characterized by "functional recovery" (repair of mitochondrial function and lipid metabolism) rather than "metabolic reprogramming."¹⁴⁷ This age-dependent difference provides a single-cell-level explanation for why the same exercise regimen can have vastly different effects in different individuals.¹⁴⁷

Based on the cross-system analysis of the three dimensions described above, and integrating published single-cell literature, we propose the following three hypotheses. First, the anti-aging effect of exercise is essentially a systemic reprogramming of the "immune-communication-metabolism" network throughout the body. The core nodes of this network include inflammatory signaling pathways (as "harmful signals"), immune cells (as "signal relays"), the circadian gene network (as the "molecular clock"), the extracellular matrix (as the "communication platform"), and various tissue stem cells (as the "effectors"). The skeletal muscle immune remodeling, stem cell niche repair, and myokine regulation observed in our study^{67,147} are a specific manifestation of this systemic reprogramming within muscle tissue. Second, immune cells serve as the central hub connecting the exercise stimulus to multi-tissue responses. Whether it is the polarization shift of macrophages in skeletal muscle, the reprogramming of microglia in the hippocampus, or the reversal of myeloid bias in HSCs in the bone marrow, all point to the central role of the immune system in mediating the systemic benefits of exercise. Immune cells not only act as effector cells that directly respond to the exercise stimulus but also act as signal relays, transmitting exercise-related information to parenchymal cells and stem cells in different organs. Third, the biological effects induced by exercise exhibit significant dose and age dependence. In terms of exercise parameters, a single bout of acute high-intensity exercise can trigger early adaptive responses such as ECM remodeling. Five weeks of moderate-term exercise is sufficient to reverse chronic inflammation and repair intercellular communication in aged individuals. In contrast, 12 months of long-term exercise can establish an anti-infection reserve in young individuals. From an age perspective, the response of aged individuals to exercise is characterized by inflammation reversal and neuroprotection, whereas the response of young individuals is dominated by metabolic optimization and the establishment of an anti-infection reserve. These findings suggest that "precision exercise prescriptions" for

different age groups require differentiated design—older adults may benefit most from moderate-duration (weeks to months) aerobic exercise, while younger individuals may require longer-term exercise interventions to build metabolic and immune reserves.

Based on the findings of this review, we propose the following suggestions for future research. First, further studies should dissect how systemic signals generated by exercise (myokines, metabolites, extracellular vesicles) are simultaneously sensed by immune cells in different organs and translated into tissue-specific responses. Second, by combining single-cell multi-omics with organ-specific gene manipulation models, researchers should establish a complete causal chain from "exercise stimulus" to "immune remodeling" to "stem cell function maintenance." Finally, based on age-dependent and exercise-dose-dependent response differences, precision exercise intervention clinical trials targeting different age groups and exercise parameters should be conducted to validate the translation of single-cell discoveries into clinical practice.

7. Limitations associated with exercise-related single-cell studies

Although single-cell omics technologies have brought breakthrough advances to exercise anti-aging research, some limitations persist in current technological applications and related studies, constraining research depth and translational efficiency. First, insufficient biological replicates - with most studies only having 1 or 2 replicates and few achieving 3 or more - and biological damage during cell isolation and library construction jointly compromise statistical power and data reliability. The single-cell isolation process itself readily induces cellular damage, leading to reduced cell viability and distortions in molecular expression, while potential damage in library construction further exacerbates such molecular expression biases.¹³³ This is particularly pronounced for dense tissues such as skeletal muscle and nervous tissue, which pose greater isolation challenges and are prone to the loss of rare cell subsets - further compromising the integrity of the study. Meanwhile, existing single-cell sequencing technologies largely rely on nucleic acid profiling following cell lysis, which hinders dynamic tracking of living cells and fails to capture the real-time changes in cellular molecular characteristics after exercise intervention. Additionally, many studies employ fluorescence-activated cell sorting to enrich specific cell populations (e.g. immune cells) for cost reduction. While this enrichment strategy lowers sequencing costs, it introduces experimental biases by excluding other cell types, potentially leading to conflicting results across studies.¹⁶¹

Second, studies at the spatial omics level remain scarce. Most current investigations focus on the gene expression profiles of single cells but lack spatial positional information, which precludes the clarification of cellular spatial distribution and intercellular interactions within the tissue microenvironment. Consequently, it is difficult to fully dissect the spatial regulatory mechanisms underlying exercise-mediated remodeling of the tissue microenvironment. Furthermore, the majority of existing studies are cross-sectional, with a dearth of long-term longitudinal tracking data. This prevents the revelation of how exercise intervention dynamically regulates cellular senescence trajectories and makes it challenging to distinguish between short-term stress responses and long-term anti-aging effects.

Finally, the translation of technological advances to clinical practice remains inadequate. Most existing studies are based on animal models or *in vitro* cell experiments, with a relative paucity of research on human samples. Moreover, the lack of standardized protocols for exercise interventions (intensity, duration, and type) hinders the translation of basic research findings into personalized clinical exercise regimens. Additionally, the molecular targets identified by single-cell omics technologies remain largely at the mechanistic and single-organ level, lacking targeted validation tools and intervention strategies. As such, a complete translational chain of "mechanism discovery → target validation → clinical application" has yet to be established.

8. Conclusions and perspectives

Aging is a systematic process involving the dysfunction of multiple organs and cell types. Traditional bulk omics studies, which fail to uncover the complex cellular heterogeneity and dynamic changes in intercellular communication networks, are unable to fully dissect the deep mechanisms of aging. In recent years, the rapid advancement of single-cell omics technologies has brought revolutionary breakthroughs to elucidating the cellular and molecular basis of aging and the mechanisms of intervention strategies at ultra-high resolution.

This review, based on single-cell omics technologies, systematically summarizes the latest progress in elucidating the mechanisms by which exercise delays aging in the skeletal muscle, nervous system, and cardiovascular system, and further dissects the systemic effects of exercise on multiple organs/tissues, as well as intercellular and inter-organ communication, from a single-cell perspective. The core conclusions are that the single-cell atlas of aging exhibits high heterogeneity. In skeletal muscle, the brain, and blood vessels, aging does not exert uniform effects on all cells, but rather specifically drives characteristic changes in the number, state, and function of cell subsets such as MuSCs, neural precursor cells, and specific endothelial cell subtypes, accompanied by the formation of a chronic inflammatory microenvironment and dysregulation of intercellular communication networks.

Conversely, exercise exerts its anti-aging effects in a cell-specific and systemic networked manner, restoring the function of specific cell types across different organs. The core mechanisms include remodeling the immune microenvironment to suppress pro-inflammatory states, repairing stem cell niches to optimize their function and differentiation potential, regulating key signaling pathways such as notch, TGF- β , and myokine signaling to restore normal intercellular communication, and reversing age-related dysregulation of metabolism and protein homeostasis.

Single-cell omics technologies provide core support for dissecting the precise mechanisms of exercise-mediated anti-aging, clearly delineate the molecular network underlying exercise-regulated multi-system aging, and lay a theoretical foundation for the development of subsequent anti-aging intervention strategies.

Although single-cell studies have achieved breakthrough progress, translating these findings into precise anti-aging intervention strategies still requires in-depth exploration in multiple directions in the future. For instance, it is essential to explore how different aged organs function systematically under the influence of exercise. Moreover, longitudinal cohort studies should be conducted in elderly populations, combining single-cell omics and clinical phenotypic data to identify cellular and molecular markers that can predict exercise responsiveness, and establish prediction models associated with clinical outcomes such as muscle strength and cognitive function.

Based on the single-cell atlas characteristics of individuals with different ages, genders, and underlying disease conditions, future research should investigate the specific effects of exercise regimens of varying types (aerobic, resistance, high-intensity interval training), intensities, and durations on specific cell subpopulations.

CRediT authorship contribution statement

Yuqi Zhao: Writing – original draft, Writing – review & editing, Visualization, Investigation, Conceptualization. **Ting Ni:** Writing – review & editing, Supervision, Funding acquisition.

Funding

This study was supported by the National Key Research and Development Program of China (2023YFC3603300, awarded to T.N.), the National Natural Science Foundation of China (32370592, awarded to T.N.) and the Shanghai Municipal Science and Technology Major Project (grant no. 2023SHZDZX02, awarded to T.N.).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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